

17 August 2026

Editors-in-Chief

Journal of Cheminformatics

Re: Submission of a Research Article

Candidate-pool opportunity and selection stability under finite validation in molecular property prediction: a repeated nested audit

Dear Editors-in-Chief,

On behalf of all authors, we are pleased to submit the Research Article entitled “*Candidate-pool opportunity and selection stability under finite validation in molecular property prediction: a repeated nested audit*” for consideration in *Journal of Cheminformatics*.

This manuscript addresses a recurrent methodological problem in molecular property prediction. Expanding a candidate registry changes both the set of available models and the difficulty of selecting among them with finite validation data. Consequently, a smaller completion gap can be misinterpreted as improved selection stability even when the apparent change is driven primarily by newly available candidates.

We introduce an exact decomposition of the full-registry completion gap into candidate-availability and within-prefix selection components. We evaluate this decomposition in a fixed 32-candidate registry across nine public molecular-property endpoints using ten split seeds and repeated 3×3 nested scaffold evaluation. Classification ROC-AUC and regression RMSE remain on their independent natural scales. The manuscript retains mixed and negative findings, including cases in which the two components move in opposite directions, and therefore avoids presenting candidate-pool expansion as uniformly beneficial.

The work is well suited to *Journal of Cheminformatics* because it provides a reproducible evaluation framework for molecular machine learning, candidate selection, and chemical support auditing. The submission includes editable figures, machine-readable source tables, split assignments, inner- and outer-level outputs, executable code, environment metadata, and cryptographic checksums to support independent verification.

We confirm that the manuscript is original, has not been published, and is not under consideration by another journal. All authors have reviewed and approved the submitted version and agree with its submission to *Journal of Cheminformatics*. The authors declare no competing interests. The study involved no human participants or animals. Any AI-assisted editorial or code-review support is disclosed in the manuscript, and the authors retain full responsibility for the scientific content and conclusions.

Thank you for considering our manuscript. We believe that its distinction between candidate opportunity and selection realization will be useful to researchers designing and reviewing molecular-property prediction studies.

Sincerely,

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